

DHIR AGRAWAL

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Summary

BCA graduate with hands-on experience building full-stack and machine learning applications, including an F1 Race Prediction System, a Customer Churn Predictor, and a Food Security Early Warning System (FoodGuard India) using Python, React, FastAPI, and Scikit-Learn. Comfortable working across the full stack — from data pipelines and ML model training to REST APIs and interactive frontend dashboards. Passionate about applying machine learning to real-world problems and actively seeking an IT role or internship to contribute to impactful projects.

Education

Veer Naramad South Gujarat University

June 2023 – May 2026

Bachelor of Computer Applications (BCA)

Surat, Gujarat

Shri C.C. Shah Sarvajanik English High School

March 2023

Higher Secondary School (Percentage: 78%)

Surat, Gujarat

Key Skills

Programming Languages: Python, JavaScript, C, SQL

Machine Learning: scikit-learn, pandas, NumPy, Matplotlib, feature engineering, cross-validation

Database: MySQL, PostgreSQL, Firebase

Frontend: React, Vite, Tailwind CSS, HTML

Backend: FastAPI, Flask, REST APIs

Tools: Git, GitHub, Google Colab, Postman

Soft Skills: Problem-solving, Teamwork, Communication, Time Management

Projects

Dhir's Pit Wall — F1 2026 Race Prediction Dashboard

Key Skills: Python, XGBoost, FastF1, FastAPI, React, Vite, Tailwind CSS

🌐 GitHub Repository 🔗 Live Demo

May 2026 – Present

- Built an end-to-end machine learning pipeline that pulls live telemetry from the official F1 API via FastF1 and engineers 13 pre-race features per driver including grid position, rolling form, pit stop speed, lap consistency, qualifying gap, track temperature, and regulation flags.
- Trained an XGBoost classifier using Leave One Race Out (LORO) cross-validation on 2026 season data, achieving an F1 score of 0.857 and AUC of 0.968 on a 66-row training set.
- Built a FastAPI backend serving live race data and predictions, with in-memory caching to avoid redundant FastF1 calls, and a React frontend displaying driver standings, race calendar, and full model transparency including feature importances and evaluation metrics.
- Validated model performance race-by-race throughout the 2026 season using real results, with an automated prediction vs actual result comparison section that renders post-race on the live dashboard.

Churno – Customer Churn Prediction System

Key Skills: Python, FastAPI, React, XGBoost, PostgreSQL, TailwindCSS

Jan 2026 – Present

- 🌐 Github Repository Link
- Developed a full-stack machine learning application to predict telecom customer churn using an XGBoost pipeline.
- Built a high-performance REST API with FastAPI to handle real-time inference and batch CSV processing.
- Designed a modern, interactive dashboard using React and Recharts to visualize customer risk distribution and trends.
- Integrated PostgreSQL for persistent history tracking and implemented an efficient bulk prediction workflow.

FoodGuard India — Food Security Early Warning System

Key Skills: Python, Scikit-Learn, Pandas, NumPy, Plotly, Streamlit

May 2026 – Present

- 🌐 Github Repository Link
- Designed a 2-step ML early warning system: trained a Random Forest to predict 3-month forward price growth and combined it with rainfall deficits and baseline nutrition into a proactive risk index.
- Engineered a monthly panel from 2.1M retail prices and daily rainfall records, resolving schema anomalies and achieving a 3.49% MAE on chronological test data (post-2023).
- Built a Streamlit dashboard with cached models and interactive Plotly charts, displaying state-level prediction trends and feature importances.

Certificate & Achievements

Supervised Machine Learning: Regression and Classification – Coursera

May 2025

Andrew Ng

Surat, Gujarat